

ASSUMPTION UNIVERSITY
VINCENT MARY SCHOOL OF ENGINEERING, SCIENCE & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE
COURSE OUTLINE

COURSE ORGANIZATION		
Course Title:	CSX3007 (Computer Architecture) sections 541, 542, and 543	
Course Status:	Major Required	
Pre-requisite:	None	
Credits:	3 Credit Points	
Semester:	2/2025	
Classroom:	VMES	
Day & Time:	Thursday (09:00– 12:00), Friday(09:00-12:00, and 13:30-16:30)	
Description:	This course will cover the following areas of computer architecture: computer system components, number system, logic gates, combinatory circuits, sequential circuits, micro-operations using flip-flop, registers, counters, arithmetic circuits, memory elements, Cache memory, computational operations and control of micro-operations, Instruction Set Architecture (ISA), Pipeline design, and Microprocessor without Interlocked Pipeline Stages CPU(MIPS- CPU) design case study.	
Objectives:	- The course's primary objective is to present, as clearly and thoroughly as possible, the nature of modern computer architecture with various computer component levels such as the internal bus system, interrupts, the internal memory system, I/O units, and the CPU. - The emphasis will be on various computer architectures and design types, covering ISA, memory architecture, and modern pipeline processors.	
Marks Allocation:	Term Project (individual)*	15 %
	Classwork (Quizzes & Assignments)	(Q1 = 7%, Q2 = 8%, Assignments = 10%)
	Midterm Examination	20 %
	Final Examination	40 %

COURSE INSTRUCTOR	
Instructor:	Asst. Prof. Dr. Anilkumar K Gopalakrishnan (office: VMES0401)
Email:	aGopalakrishnan@au.edu
Course materials:	MS Teams
Mobile Phone:	0891351711

COURSE RESOURCES	
Textbooks:	Digital Design and Computer Architecture, David M.H and Sarah L.H, Morgan Kaufmann, Elsevier (ISBN: 978-0-12-394424-5)
Reference(s):	- Computer Architecture and Implementation, Harvey G. Cragon, Cambridge University Press (ISBN: 0-52-165168-9) - Computer Architecture: A Quantitative Approach, David A. Patterson and John L. Hennessy, Morgan Kaufmann Publishers, (ISBN: 0-12-370490-1)

COURSE POLICIES:
1. Students must have at least 80% class attendance to be eligible for the final examination.
2. If a student is absent from more than three classes <u>without any reason</u> , then they should withdraw from the course immediately.

ASSESSMENT APPEAL POLICY
For any assignments/projects and examination(s) (excluding the final examination), the lecturer will announce scores and discuss students' solutions approximately within 1-3 weeks after the submission deadline and after grading is complete. Students may request an assessment appeal from the lecturer within one week or by the specified deadline.

COURSE CONTENTS AND TENTATIVE SCHEDULE		
Week	Topic	Remarks
Week 1,2	Introduction-Number System & Logic Gates	Chapter 1
Week 3, 4	Combinational Logic Design	Chapter 2
Week 5	Sequential Logic Design	Chapter 3
Week 6,7	Arithmetic Circuits, Sequential Building Blocks,	Chapter 5
	Quiz 1	
	Midterm Exam	
Week 8	Sequential Building Blocks	Chapter 5
Week 9	Instruction Set Design (ISA)	Chapter 6
Week 10	Instruction Set Design, Performance Evaluation	Chapter 6
Week 11 & 12	MIPS Pipeline Architecture	Chapter 7
Week 13	Project Presentation	
Week 14	Quiz II	
Week 15	Final Exam	

TERM PROJECT DESCRIPTION:

Write a simple Instruction Set (ISA) simulation for a **32-bit CPU** using any programming language. The ISA should support **arithmetic operations**, including *addition*, *subtraction*, *multiplication*, and *division*. Additionally, the ISA must include **CPI (Clock per Instruction)**.

Requirements:

- The simulation must support a 32-bit integer number or more.
- The ISA should consist of at least eight general-purpose registers (GPRs), each 32 bits in size.
- The program should be able to handle (at least) op-codes such as *mov* (move), *add* (addition), *sub* (subtraction), *mul* (multiplication), *div* (division), etc.

Example ISA

A 24-bit ISA with 16-bit integer data representation and 16-bit 8 GPRs (say r0,..., r7) is demonstrated here. The register **r7** from the GPR acts as a **multiplication register** for multiplication and a **remainder register** for division (avoid using **r7** as a source register for multiplication and division). An example instruction set with its decoded and encoded form is shown below:

<u>Decoded form</u> of the <u>Instruction meaning</u> .	<u>Encoded form</u>
mov r1 3 r1 ← 3	[0001001 0000 0000 0000 0011]
add r1 3 r1 ← r1 + 3	-----
mov r2 r1 r2 ← r1	-----
mul r2 -1 r7: r2 ← r2 * -1 (total 32-bit result)	-----
mov r3 r2 r3 ← r2	-----
div r3 2 r7:r3 ← r3 / 2	-----
end 0 0	

Values of registers after the execution of the instruction set

During the execution of the above code sequence, the values of registers would be varied in the following way:

r1 = 3 [0000 0000 0000 0011]
r1 = 6 [0000 0000 0000 0110]
r2 = 6 [0000 0000 0000 0110]
r7: r2 = -6 [1111 1111 1111 1111 1111 1111 1010]
r3 = -3 [1111 1111 1111 1101] r7: 0 [0000 0000 0000 0000]

CPI (Clocks Per Instruction)

The CPI of the program depends on the number of clock cycles each instruction requires.

Pipelined version

The pipelined version of the instruction set simulation is an optional part of the project. But, whoever With such an additional feature, you will be significantly rewarded.

Scoring Criteria:

The following key qualities will be considered:

- ISA design & coding 10%
 - Project idea with code (report) 5%
- *Submitting a hard copy project report is compulsory.**